WEEK 2 · DAY 5 · OVERVIEW

Case Study

Convexity and the September Book



Created by **Jeff Lenamon**

ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

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Problem Definition

Notebook 09 put our four positions at 6,154.97 dollars a basis point: a straight line. For a 100 basis point move, how far off is that line, and what measures the bend? Put that measure in bondmath, tied to Excel, which has no function for it. Then September's holdings: 200 bonds priced by someone else's program. Rebuild every bond's price, accrued, yield and duration before we use the file, and run the same checks on the benchmark it came from. Last, what does the book carry in dollars per basis point, and what is its duration? Describe the book: no view on any bond.

ABOUT 45 MIN WITH EXERCISES / 10 SECTIONS / 4 EXERCISES

Your Route Through Day 5

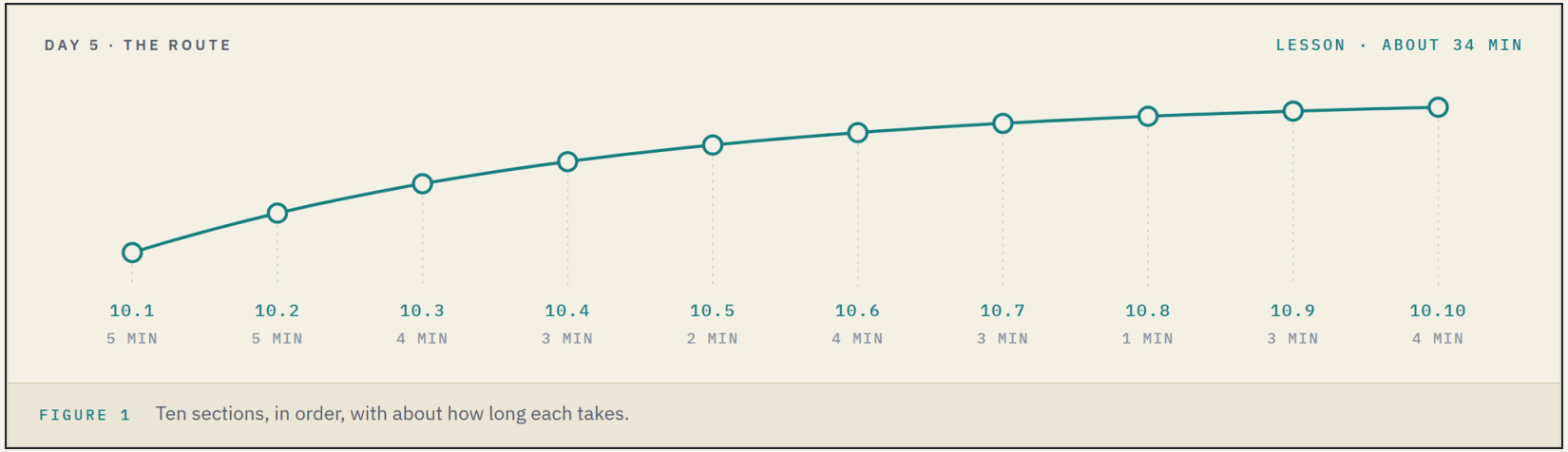


FIGURE 1 Ten sections, in order, with about how long each takes.

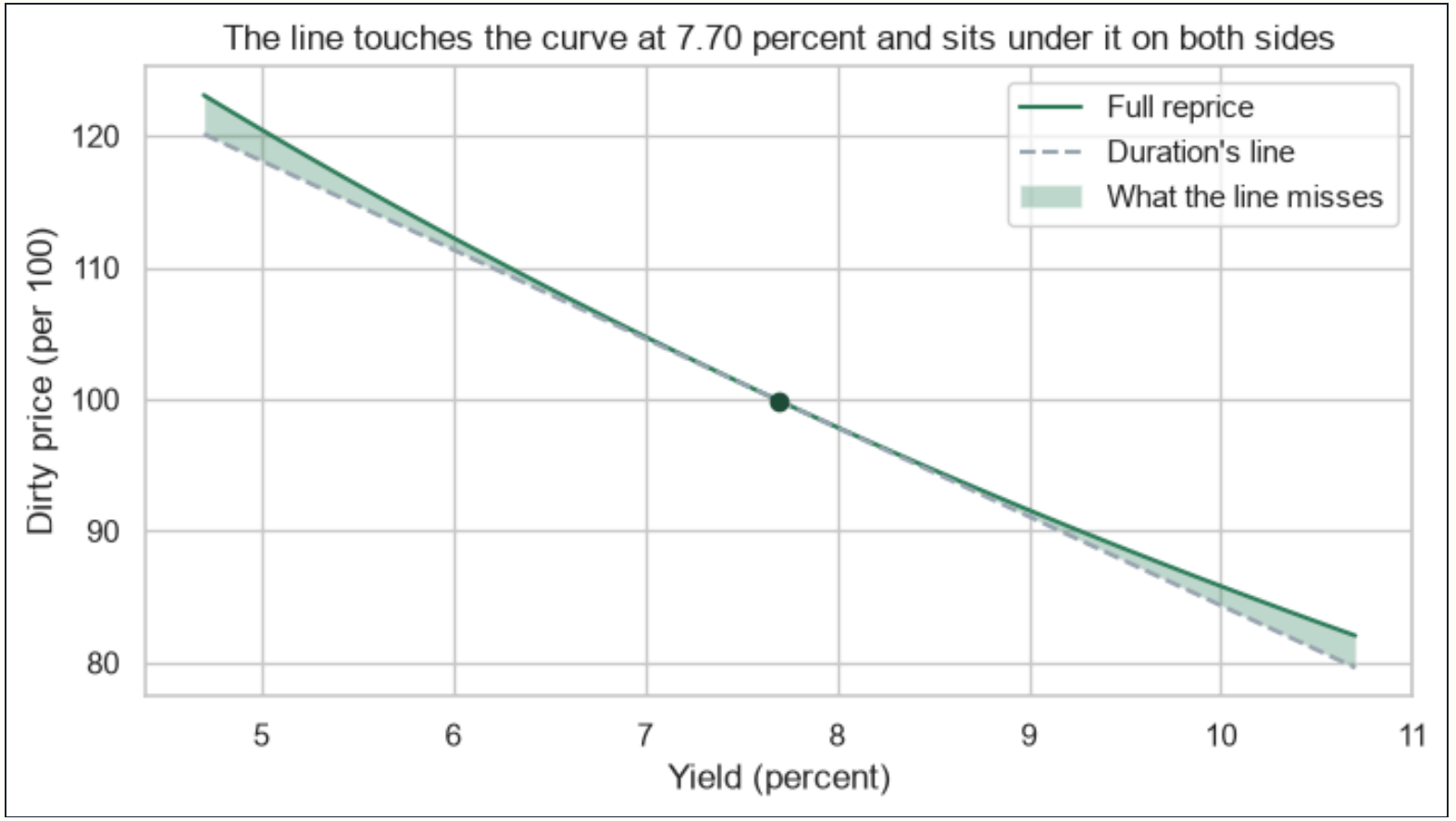
	SECTION	ABOUT
10.1	Where Duration's Straight Line Misses	5 min
10.2	Convexity and **convexity**	5 min
10.3	Checking It When Excel Has No Convexity Function	4 min
10.4	Duration Plus Convexity Against a Full Reprice	3 min
10.5	Tag the Week	2 min

	SECTION	ABOUT
10.6	The Book's Price and Accrued	4 min
10.7	Yield From the File's Price	3 min
10.8	Duration, and Convexity as a New Column	1 min
10.9	Two Systems Disagree, Again: the Benchmark	3 min
10.10	The Book in Dollars per Basis Point	4 min

Time. About 34 minutes for the lesson, then about 12 for the four exercises.

10.1 Where Duration's Straight Line Misses

CURVE AND LINE



DURATION'S LINE FOR 100 BP

```
The line for +100 bp: -6.7511 per 100
```

10.2 Convexity and convexity

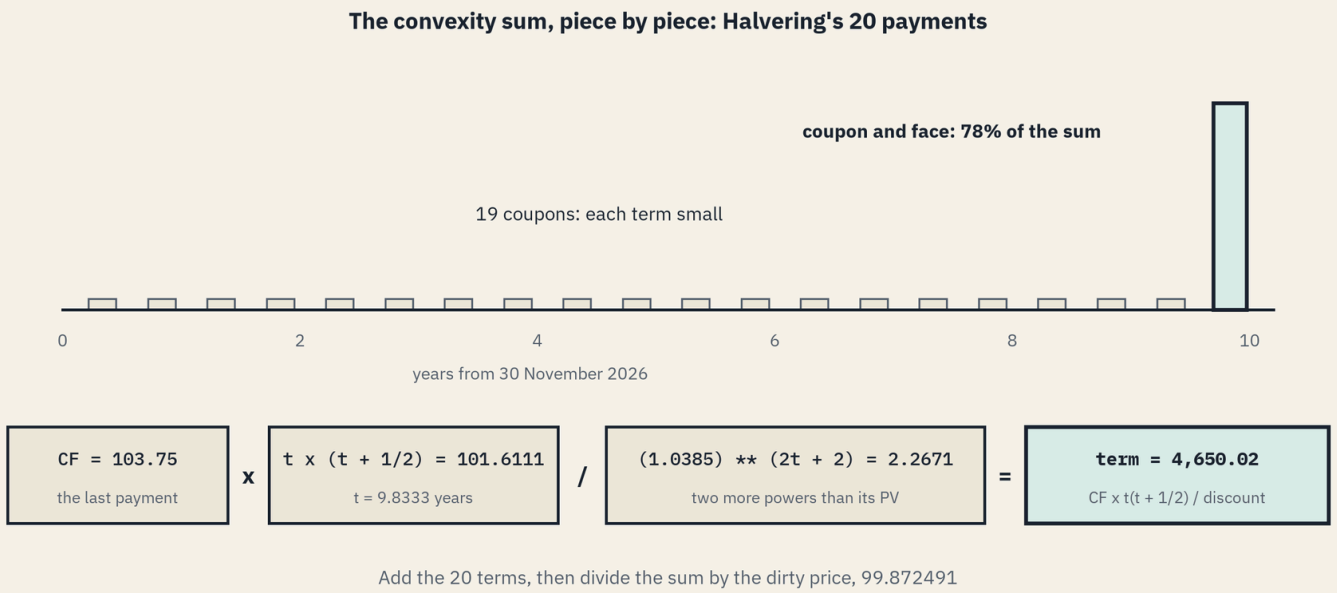


FIGURE 2 Halvering's last payment adds a term of 4,650.02, and the 20 terms sum to its convexity.

THREE SCALINGS

```
Unscaled 59.5474, divided by 100 0.5955, halved 29.7737
```

This course's is unscaled: ask which scaling any other number uses before comparing.

10.3 Checking It When Excel Has No Convexity Function

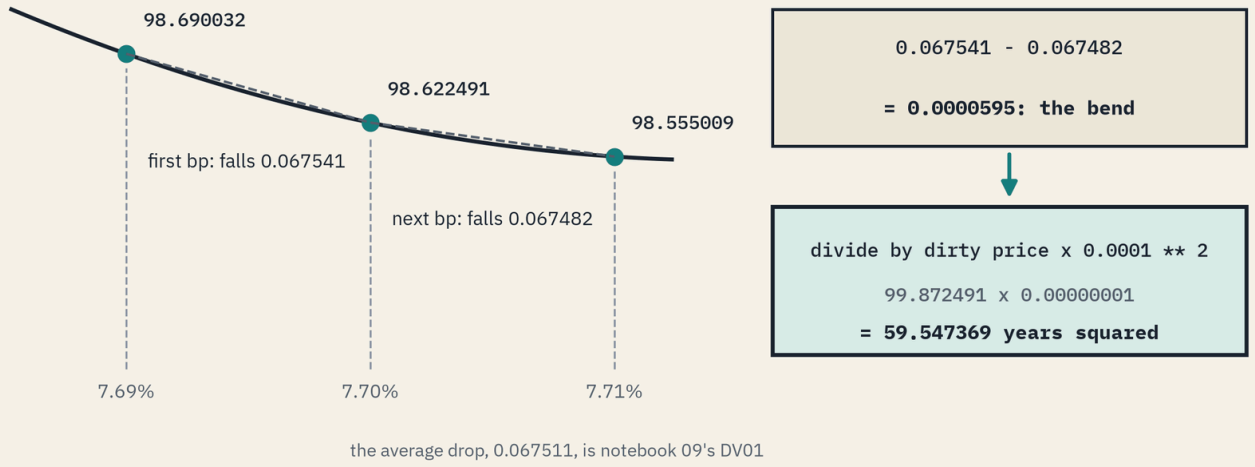


FIGURE 3 Prices 98.690032, 98.622491 and 98.555009 at 1 bp steps give a bend of 59.547369 years squared.

CENTRAL DIFFERENCE

```
Central difference 59.547369, convexity 59.547364
Gap 4.8e-06
```

Excel's Three PRICE Cells

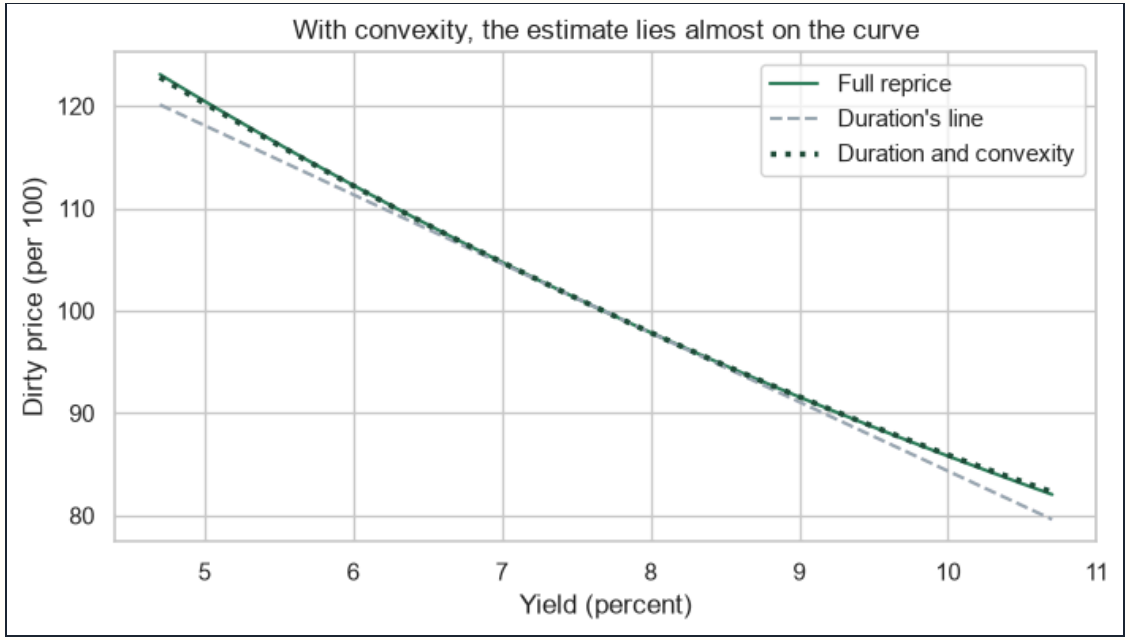
EXCEL SIDE BY SIDE

JOB	EXCEL	RESULT
Clean price, A1	`=PRICE(DATE(2026,11,30),DATE(2036,9,30),7.5/100,7.7/100,100,2,0)`	98.622491
1 bp lower, A2	`=PRICE(DATE(2026,11,30),DATE(2036,9,30),7.5/100,7.7/100-0.0001,100,2,0)`	98.690032
1 bp higher, A3	`=PRICE(DATE(2026,11,30),DATE(2036,9,30),7.5/100,7.7/100+0.0001,100,2,0)`	98.555009
Central difference, A5	`=(A2+A3-2*A1)/((A1+A4)*0.0001^2)`	59.54736851639378

`convexity` **matches Excel's three** `PRICE` **cells on all 294 rows outside the final period.**

10.4 Duration Plus Convexity Against a Full Reprice

CURVE, LINE, AND PARABOLA



FOUR BONDS AT +100 BP

	ISSUER	ESTIMATE_PCT	FULL_PCT	MISS_PCT
0	Quorvane Energy	-4.075935	-4.077845	-0.001910
1	Dunmarrow Rail	-5.280391	-5.284763	-0.004372
2	Pellwick Paper	-2.563475	-2.563985	-0.000510
3	Halvering Foods	-6.462029	-6.471508	-0.009478

Two terms close the miss from 0.29 to 0.01 per 100 at 100 bp: the note's first two asks.

10.5 Tag the Week

READ THE DIFF, COMMIT

```
!git -C "{work}" status --short
!git -C "{work}" diff --stat
```

```
M bondmath.py
 M test_bondmath.py
 bondmath.py      | 30 ++++++++++++++++++++++++++++++
 test_bondmath.py |  9 +++++++++
```

NAME THE WEEK

```
!git -C "{work}" tag week2
!git -C "{work}" log --oneline --decorate
```

```
646a02a (HEAD -> main, tag: week2) Add convexity, checked against a central difference on Excel PRICE
5fcb7c6 Notebook 09: macaulay_duration, modified_duration, and dv01
```

`convexity` is week 2's last function: commit it, and name the commit that ends the week.

10.6 The Book's Price and Accrued

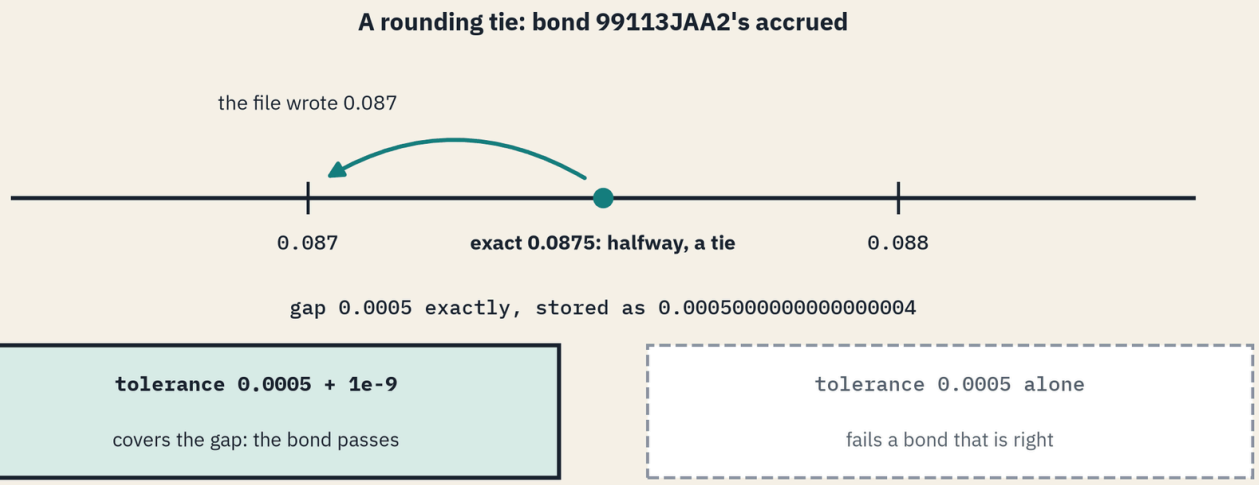


FIGURE 4 Bond 99113JAA2's exact accrued of 0.0875 is a tie, so the file wrote 0.087.

THE WIDEST GAP

```
cusip            99113JAA2
coupon                5.25
accrued_calc        0.0875
accrued              0.087
Name: 158, dtype: object
```

Price and accrued tie on all 200 bonds; the one gap a hair above 0.0005 is a rounding tie.

10.7 Yield From the File's Price

TWO TOLERANCES, WRITTEN ONCE

```
ROUNDED_3 = 0.0005 + 1e-9
YIELD_FROM_ROUNDED_PRICE = 0.001
```

SOLVE 200 YIELDS

```
book['yield_calc'] = [bondmath.yield_to_maturity(s, m, c, p) for s, m, c, p
book['yield_gap'] = (book['yield_calc'] - book['yield_pct']).abs()
assert book['yield_gap'].max() <= YIELD_FROM_ROUNDED_PRICE
```

Now the other way: solve each yield from the file's rounded price with notebook 08's `yield_to_maturity`.

10.8 Duration, and Convexity as a New Column

DURATION, AND A NEW COLUMN

```
duration_gap = (book['duration_calc'] - book['duration']).abs()
print(f'Largest duration gap {duration_gap.max():.7f} years')
book['convexity'] = [bondmath.convexity(*bond) for bond in inputs]
```

```
Largest duration gap 0.0004982 years
```

AGAINST EXCEL'S DIFFERENCE

```
excel_fd = reference[reference['group'] == 'holdings'].set_index('case_id')['convexity_fd']
convexity_gap = (book.set_index('cusip')['convexity'] - excel_fd).abs()
assert len(convexity_gap) == 200 and convexity_gap.max() < 1e-3
print(f'Largest gap {convexity_gap.max():.1e}, on {convexity_gap.idxmax()}')
```

```
Largest gap 2.2e-04, on 99053BAB9
```

All four columns rebuilt, and a fifth added that Excel confirms: the note's third ask.

10.9 Two Systems Disagree, Again: the Benchmark

ONE FUNCTION FOR ANY FILE

```
def column_gaps(file):
    rebuilt = pd.DataFrame({
        'price': [bondmath.price(*bond) for bond in terms],
                  in zip(settle, maturity, file['coupon'], file['price'])],
    return (rebuilt - file.set_index('cusip')[rebuilt.columns]).abs()
```

821 BONDS

```
benchmark = pd.read_csv('benchmark.csv')
benchmark_gaps = column_gaps(benchmark)
outside = benchmark_gaps[(benchmark_gaps > tolerance).any(axis=1)]
print(len(benchmark_gaps), 'benchmark bonds,', len(outside), 'outside a tolerance')
outside.round(6)
```

```
821 benchmark bonds, 2 outside a tolerance
```

819 of 821 tie; the two that do not are notebook 06's February month-end bonds, a convention, not an error.

10.10 The Book in Dollars per Basis Point

WHAT IT PRINTS

```
Market value 490,819,215.00 dollars
Book DV01 306,996.11 dollars per bp, duration 6.2548 years
```

EXCEL SIDE BY SIDE

JOB	PYTHON	EXCEL	RESULT
Book DV01	`book_dv01_dollars`	`=SUMPRODUCT(E2:E201,H2:H201)/100`	306,996.11
Market value	`total_market_value`	`=SUM(I2:I201)`	490,819,215.00
Book duration	`book_duration`	`=SUMPRODUCT(I2:I201,J2:J201)/SUM(I2:I201)`	6.254772

The book moves 306,996.11 dollars a basis point, duration 6.2548 years: the note is answered.

You Can Now

- Say where duration's line misses a full reprice, and why the miss grows with the square of the move.
- Compute convexity with **`bondmath.convexity`** and say how other sources scale it.
- Check convexity against three **`PRICE`** cells, since Excel has no convexity function.
- Estimate a price change with duration and convexity.
- Rebuild a file's four columns at stated tolerances, and leave out a known exception by name.
- Add up a book's DV01 in dollars and its weighted duration, and say why they are one number.
- Compare two tags with **`git diff --stat`** and **`git log`**.

NEXT
WEEK 3 · DAY 1

Treasury Curve Interpolation

A yield at any maturity from a curve published at a few tenors: a straight line between two of them, interpolate_yield in the module, tied to Excel, and what the straight line misses.